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Augmented methanogenic performance in straw-manure co-digestion via micro/nano bubble-enhanced syntrophic metabolism

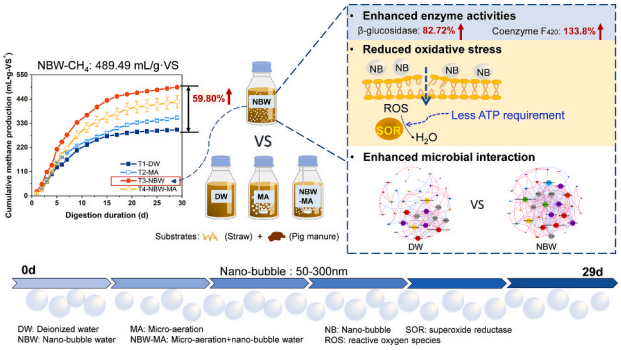
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HIGHLIGHTS

- NBW treatment yielded the highest methane production over other aeration methods.
- Enzyme activities and microbial diversity were enhanced in NBW treatment.
- Less energy was spent to cope with the ROS oxidation stress in the NBW treatment.
- Methanosaeta (91.66 %) were key archaea for methane production.

GRAPHIC ABSTRACT



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ABSTRACT

This study aimed to investigate the enhancement of methane production by comparing air-nanobubble water (air-NBW) and conventional micro-bubble aeration during the anaerobic co-digestion (AcoD) of rice straw and pig manure. Remarkably, the addition of air-NBW (T3-NBW), as the most effective gas supplementation strategy, resulted in the highest cumulative methane yield of 489.49 mL/g·VS with a minimal lag phase of 0.57 days, representing a 59.80 % increase over the control (306.49 mL/g·VS, $p < 0.001$). This finding aligns with the kinetic fitting data (477.85 ± 5.84 mL/g·VS), wherein the maximum methane production rate for the T3-NBW was 41.89 mL/g•VS·d^{-1}. Mechanistic insights revealed that air-NBW dramatically elevated the activity of β-glucosidase (hydrolysis) and coenzyme F_{420} (methanogenesis) by 82.72% and 133.8 %, respectively, compared to the control. These results suggested that the efficient conversion of intermediate products played a critical role in enhancing overall methane yield. Microbial community analysis identified acetotrophic methanogenesis as the dominant pathway, with *Methanosaeta* abundance reaching 91.66 % in NBW-amended treatments. Additionally, the co-occurrence network analysis showed that the air-NBW system promoted microbial interaction and stabilized the ecological network. Crucially, the NBW treatment exhibited the lowest expression of energy-dependent antioxidant enzyme genes (e.g., superoxide reductase, *SOR*), suggesting reduced oxidative stress and enhanced energy allocation toward microbial growth and methanogenic activity. By enhancing oxygen

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dispersion and microbial intracellular redox balance, NBW technology provides a promising approach for improving methane recovery in sustainable waste-to-energy systems.

1. Introduction

As global energy shortages and environmental pollution issues intensify, the demand for sustainable and renewable resources is experiencing rapid growth (Khan et al., 2021). Biomass, like rice straw and pig manure, is recognized as a renewable energy source. The annual production of rice straw and pig manure are approximately 900 million tons and 700 million tons in China, respectively (Chang et al., 2021; Wang et al., 2022). However, most of these waste streams are not effectively treated and utilized, which leads to biomass resource loss but also pollutes the environment (Im et al., 2019; Yuan et al., 2017). Biogas production from anaerobic digestion (AD) provides a promising approach to mitigating environmental pollution but also alleviating energy shortages (Li et al., 2024).

Recently, anaerobic co-digestion (AcoD) has been widely used to enhance biogas production from biomass. Generally, rice straw features in high carbon/nitrogen (C/N) ratio and rich lignocellulosic content while pig manures are characterized by a lower C/N ratio. Direct anaerobic fermentation of these individual substrates often leads to excessive accumulation of organic acids or ammonium, resulting in the failure of the anaerobic system. The AcoD of rice straw and pig manure provides a better macro- and micro-nutrient balance, pH buffering capacity, and process efficiency (Wang et al., 2018). Nevertheless, recalcitrant lignocellulosic structure limits the hydrolysis and mass transfer rate during the co-digestion of straw and pig manure, which remains a great challenge for methane production. Strategies like biochar addition and thermal pretreatment were found to be able to facilitate this progress (Chuenchart et al., 2021; Li et al., 2023). Moreover, micro-aeration technology, introducing air-microbubbles (MBs, with diameters of 1–100 μm) or nanobubbles (NBs, with diameters of <1 μm) into the fermentation liquid, has recently emerged as an effective way to enhance the performance of the AD process (Zhou et al., 2022). It was reported that limited oxygen content can increase microbial diversity and promote the enrichment of facultative hydrolytic bacteria, thus accelerating the biodegradation of organic matter (Ding et al., 2024).

Nanobubble water (NBW) has attracted significant attention in the AD process due to its unique properties, including long-term stability, negative surface charge, and the ability to generate reactive oxygen species (ROS) (Hou et al., 2024). The NBs can be maintained in water for extended periods because they are almost unaffected by buoyancy in solution. Previous studies have shown that the bubble (~140 nm) in air-NBW remained for 14 days, maintaining a stable micro-oxygen environment over the long term, thereby enhancing methane production (Wang et al., 2019). In addition, NBs show a high microbial adhesion due to the characteristics of negative surface charge and high zeta potential (Hou et al., 2024). The large specific surface could ensure sufficient contact between microorganisms and substrates, further accelerating the biochemical reaction. Research has demonstrated that NBW accelerated the consumption of volatile fatty acids (VFAs), avoiding their accumulation and associated inhibitory effects, thereby strengthening the syntrophic metabolism (Fan et al., 2024). Moreover, air-NBW was reported to be able to significantly enhance electron transfer under high salinity conditions, leading to a 21–65 % increase in hydrogen yield and a 14–43 % increase in methane yield (Hou et al., 2021a). Furthermore, NBs have been proven to generate ROS: when nanobubbles collapse, the rapid release of internal pressure generates substantial surface energy, facilitating the conversion of oxygen molecules into ROS, including hydroxyl radicals (•OH), superoxide anions ($O_2^{\bullet-}$), and hydrogen peroxide (H_2O_2) (Hou et al., 2024). On one hand, ROS was considered to be beneficial for facilitating the degradation of complex organic compounds. For instance, •OH possesses robust oxidizing properties that can readily cleave C—O and C—C bonds in lignin, leading to lignin depolymerization (Guengerich & Yoshimoto, 2018). This accelerates cellulose degradation and increases soluble organic matter content, thus resulting in an elevated cumulative H_2 yield (Zhu et al., 2023b). However, some studies indicated that excessive ROS can disrupt membrane integrity, impair metabolic pathways, and ultimately compromise the overall efficiency of the digestion process (Liu et al., 2016b).

Traditional microbubble aeration has been demonstrated as an effective strategy to enhance the process efficiency during the AD process (Fu et al., 2022). Microbubbles usually dissolve and collapse more rapidly, resulting in lower gas transfer efficiency and an increased propensity for gas leakage during aeration. This not only complicates the AD process but also poses a potential explosion risk (Chuenchart et al., 2021). In contrast, NBs distinctly improve gas dispersion and gas–liquid mass transfer, while simultaneously reducing energy consumption (Zhou et al., 2022). Moreover, there are still some controversies on the ability of ROS generation of MBs/NBs. Some studies suggested that MBs are more prone to rupture, leading to larger amounts of ROS generation than NBs (Ohgaki et al., 2010). However, some researchers insisted that NBs were more conducive to the production of •OH. The amount of exogenous •OH has a positive relationship with the NB number density in the water (Liu et al., 2016a). Therefore, a systematic comparative study on the impact of MBs/NBs on AcoD performance and the microbial response is needed.

In this study, microbubble-aeration and air-nanobubble water were proposed to enhance the performance of the AcoD of rice straw and pig manure. The impact of different aeration strategies on the hydrolysis-acidification process and methanogenesis was discussed. Furthermore, a comprehensive investigation of the microbial community structure, co-occurrence networks, key enzyme activities, and functional gene expression related to antioxidant defense systems based on the Kyoto Encyclopedia of Genes and Genomes (KEGG) database was performed to reveal the possible metabolic pathway and underlying mechanism. These findings are expected to provide new insight into more efficient methane production from biomass and contribute to the development of the circular economy.

2. Materials and methods

2.1. Substrates and inoculum

Rice straw and pig manure were used as substrates in this study. Rice straw and pig manure were collected from the experimental field and the livestock and poultry breeding center of Hunan Agricultural University, respectively. Before the experiment, the rice straw was first cut into small pieces and then sieved through a 20-mesh sieve. After removing undigested feed, hair, and large impurities, the collected pig manure was thoroughly mixed. The inoculum used in the experiment was anaerobic granular sludge purchased from Qingdao Liboyuan Environmental Protection Materials Co., Ltd. (Shandong). The sludge was pre-cultured in a thermostatic chamber at 37 ± 1 °C for one week. The main

Table 1
Characteristics of rice straw and pig manure used in this study.

Parameters	Unit	Rice straw	Pig manure	Inoculum
Total solids (TS)	%	88.24 ± 0.4	18.48 ± 0.62	10.05 ± 0.3
Volatile solids (VS)	%, of TS	84.70 ± 0.13	77.16 ± 0.71	76.71 ± 0.2
C	%	38.1	37.15	0.86
N	%	0.66	2.08	0.07
C/N	–	57.69	17.86	12.72

characteristics of the rice straw, pig manure, and inoculum are shown in Table 1.

2.2. Preparation and characterization of microbubble aeration and NBW

The air-NBW used in this experiment was prepared by North China Electric Power University (Nanolf1500 series, Shanghai Xingheng Technology Co., Ltd.). The concentration and size distribution of the NBs were analyzed and measured using a nanoparticle tracking analyzer (NS300, Malvern). The Zeta potential was measured using a Zeta potential analyzer (Zetasizer Nano ZS90). MBs are generated through a gas diffuser connected to a micron-sized aeration head, driven by a peristaltic pump (LongerPump BT100-2J) with a head capacity of Longer YZ1515x.

2.3. Experimental setup

Serum bottles with a total volume of 500 mL were used as the AD reactor. The substrates were a mixture of pig manure and rice straw with a wet weight ratio of 2:1. The initial TS concentration was set at 6.7 %. Four treatments were set: control (T1-DW), air microbubble-aeration (T2-MA), air-NBW (T3-NBW), and the combination of air-NBW and air-microbubble-aeration (T4-NBW-MA). In this experiment, 320 mL of deionized water was added to the T1-DW and T2-MA treatments, while 320 mL of air-NBW was added to the T3-NBW and T4-NBW-MA treatments. The fermentation experiment lasted for 29 days. Samples were collected every day in the first 9 days and afterwards, samples were taken every two days. After each sampling, the T1-DW and T2-MA treatments were supplemented with an equal volume of deionized water, while the T3-NBW and T4-NBW-MA treatments were supplemented with an equal volume of air-NBW. Additionally, to maintain the presence of microbubbles, 150 mL of ambient air was continuously injected into the T2-MA and T4-NBW-MA treatments at a flow rate of 0.83 mL/min after each sampling. All the tests were carried out in triplicate.

2.4. Analysis methods

2.4.1. General parameters

The C/N ratio contents were determined using an elemental analyzer (German element Elementar, MACRO cube). Total solids (TS) and volatile solids (VS) were measured according to standard methods. Soluble carbohydrate concentration was measured by using the Phenol-sulfuric method. Soluble protein concentration was determined with the Coomassie Brilliant Blue method. The volume of biogas produced during the experiment was measured using a medical syringe method. The composition of the biogas (CH_4, CO_2, H_2) was analyzed using a gas chromatograph (LC-20A, Shimadzu, Japan). The digested samples were centrifuged, and the supernatant was filtered through a 0.22 μm membrane for VFA analysis using high-performance liquid chromatography (DJS-1C, LeiMagnet, Shanghai, China); meanwhile, chemical oxygen demand (COD) was determined by the potassium dichromate method. The activities of β-glucosidase and coenzyme F_{420} were measured using UV spectrophotometry. The collected samples were centrifuged, and the pellets were used for microbial community analysis by Novogene Co., Ltd. The co-occurrence network analysis was performed using R-Project (version 4.4.2), and the resulting network was visualized using Gephi. The relative abundances of bacteria and archaea were determined via 16S rRNA sequencing. The functional gene expression related to the dominated metabolic pathway and antioxidant defense was analyzed using the KEGG database.

2.4.2. Kinetic models

The modified Gompertz model was used to study the methane production process. This model was employed to calculate the maximum methane production potential, the maximum methane production rate, and the lag phase. The model is shown in Eq. (1).

$$P(t) = P_0 \times \exp\left\{ -\exp\left[\frac{R_m e}{P_0}(\lambda - t) + 1\right]\right\} \quad (1)$$

where $P(t)$ is the cumulative methane production (mL/g·VS); P_0 is the maximum methane potential production (mL/g·VS); R_m is the maximum methane production rate (mL/g·VS·d^{-1}); λ is the time of the delay phase (d); and t is the time of AD (d).

2.4.3. Statistical analysis

Origin 2018 was employed for plotting all data in this study. Single-factor analysis of variance (ANOVA) was conducted using SPSS version 26.0 to test the significance of differences, with a significance level assumed at $p \leq 0.05$.

3. Results and discussion

3.1. Properties of air-nanobubble water

Air-NBW was generated by introducing air into deionized water using a nanobubble generator. As illustrated in Fig. 1(a–b), the air-NBW exhibited a narrower size distribution of 50–300 nm compared to the deionized water (50–500 nm). The size distribution of air-NBW in this study was a little different from the previous research on air-NBWs (144–223 nm) (Wang & Yang, 2019). The slight variations in bubble distribution can be attributed to disparities in nanobubble generation equipment and process-specific conditions, such as gas injection parameters and water quality. Furthermore, the size distribution curve of the air-NBW remained nearly constant (from 15:21:15 to 15:26:09) while the bubble size distribution of ordinary deionized water showed a significant variation over the monitoring period (from 14:40:23 to 14:44:58). This indicates that air-NBW exhibited greater stability over time compared with the deionized water.

The zeta potential of the prepared air-NBW and deionized water was also measured (Fig. 1c). Air-NBW exhibited a significantly more negative zeta potential (-18.7 mV) than deionized water (-7.14 mV). The zeta potential value was close to the reported value (-20 mV ~ -17 mV) measured in the water after the introduction of NBs (Ushikubo et al., 2010). This high negative zeta potential was attributed to the adsorption of hydroxide ions on the NB interface, forming an electrical double layer of ions and counter ions (Chuenchart et al., 2021). The increased absolute value of the zeta potential implied stronger interfacial charge density and enhanced electrostatic repulsion, contributing to higher colloidal stability. In summary, air-NBW had prolonged the gas–liquid interfacial behavior within the system, thereby creating a more favorable microenvironment for gas transfer, enabling sustained oxygen supply, and promoting substrate degradation and microbial metabolism, which was consistent with the findings in Section 3.3.

Air-NBW demonstrated distinct physicochemical properties, characterized by a bubble size distribution ranging from 50 to 300 nm, the negatively charged, and elevated zeta potential in comparison to deionized water. These features increased specific surface area and enhanced substrate–microbe interactions, accelerating biochemical reactions during AD. Previous research has shown that NBs improved gas transfer and ozone dissolution in viscous systems (Hu & Xia, 2018). In addition, NB collapse generated ROS, including •OH radicals, which facilitated substrate degradation but could be harmful to microbes if overly abundant (Liu et al., 2016b). Overall, it was expected to optimize the AD process by stimulating microbial metabolic activity with air-NBW supplementation.

3.2. Effect of micro/nanobubbles aeration on methane production

The cumulative and daily methane production profiles under different strategies were illustrated in Fig. 2. It was found that the

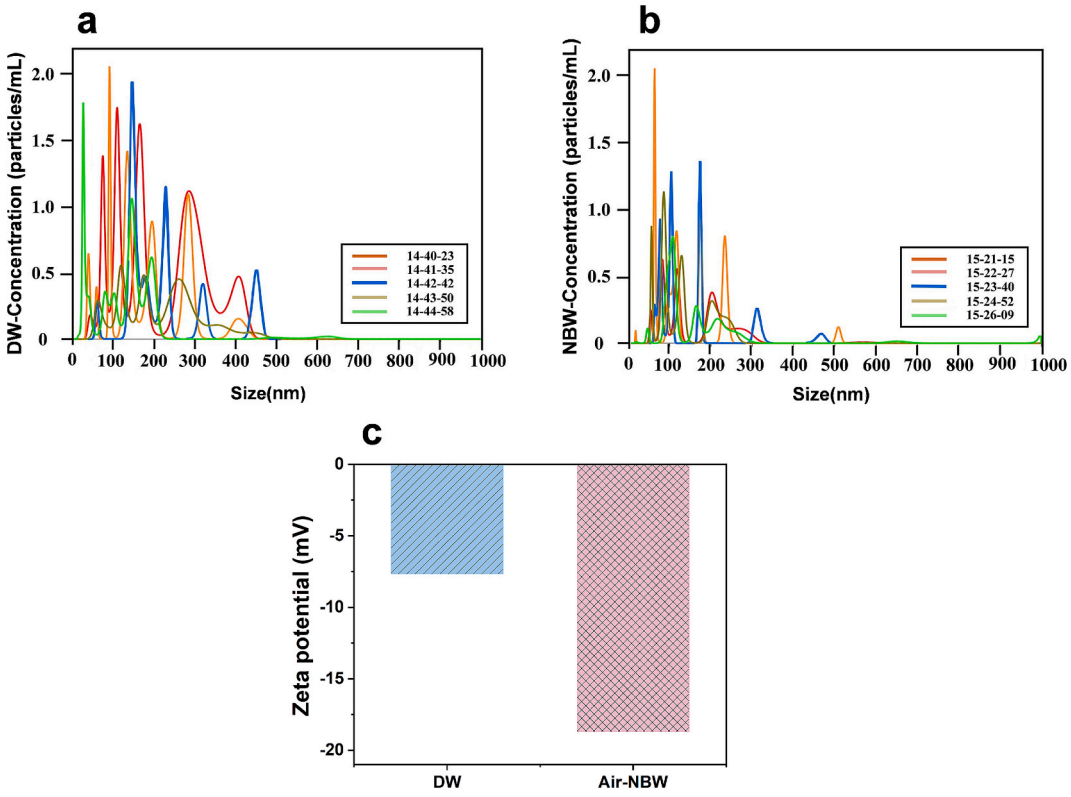


Fig. 1. Size distribution of deionized water (a) and air-NBs (b), and zeta potential of deionized water and air-NBW (c).

treatments (T2-MA, T3-NBW, T4-NBW-MA) exhibited significantly higher cumulative methane production (351.94–489.49 mL/g·VS) compared to the control (T1-DW, 306.49 mL/g·VS) (Fig. 2a). This indicated that micro-aeration strategies significantly promoted the performance of methanogenesis during AD. Among these, the T3-NBW treatment demonstrated the highest cumulative methane yield of 489.49 mL/g·VS, representing a 59.80 % enhancement over the control. The T4-NBW-MA and T2-MA treatments achieved cumulative methane productions of 408.74 mL/g·VS and 351.94 mL/g·VS, corresponding to 33.36 % and 14.83 % increases compared to the control, respectively. The highest methane yield observed in this study was significantly higher (67.67 %) than that reported value (291.77 mL/g·VS) in other studies involving the co-digestion of straw and pig manure (Han & Ning, 2024). The significant enhancement in methane production observed in this study was attributed to the addition of air-NBW which enhanced the degradation efficiency of organic matter and stimulated microbial activity. Furthermore, variations in substrate composition and operational conditions likely alleviated the accumulation of VFAs, thereby optimizing the micro-environment. The daily methane production profile (Fig. 2b) revealed that massive methane production occurred during days 1–15 for all treatments, coinciding with the decomposition of macromolecular into VFAs. Minor methane was produced afterward. Notably, the T3-NBW treatment reached peak daily methane production (70.59 mL/g·VS·d^{-1}) on day 5, demonstrating an improvement of 108.96 % over the control (33.78 mL/g·VS·d^{-1}). This enhancement observed during the start-up period may be attributed to the effect of NBs on the accelerated breakdown of complex organics, thereby rapidly releasing bioavailable VFAs (Chuenchart et al., 2021). Notably, the cumulative methane production of T3-NBW treatment reached 310.37 mL/g·VS during days 2–9, accounting for 63.4 % of the total methane production. These results demonstrated that the T3-NBW treatment outperformed other aeration methods (microbubble aeration and the combination of microbubble-nanobubble aeration) in the methane production from the AcoD of pig manure and straws. In summary, the addition of NBW enhanced methane production in the AD of rice straw and pig manure without causing secondary pollution.

In this study, the kinetic behavior of methane production during the anaerobic co-digestion of pig manure and straw was systematically evaluated using the modified Gompertz model (Fig. 2c–f). Notably, the T3-NBW treatment exhibited superior methanogenic performance, achieving a maximum cumulative methane production of 477.85 ± 5.84 mL/g·VS, which represented a 58.45 % enhancement compared to the control (301.56 ± 3.24 mL/g·VS). This result closely aligns with the actual methane yield measured in this study (489.49 mL/g·VS), demonstrating that the model accurately described the variation in cumulative methane production ($R^2 > 0.9846$). Previous research reported a predicted maximum cumulative methane production of 335.25 mL/g·VS by the Gompertz model under similar experimental conditions (initial TS content of ~6 %) (Tian & Gong, 2023), which was obviously lower than the value observed in this study. Building on this once again demonstrated the superiority of air-NBW in enhancing methane production during co-digestion. Furthermore, the kinetic analysis revealed two critical advantages of NBW supplementation: (1) NBW enhanced the methane production rate: the maximum methane production rate for the T3-NBW treatment reached 41.89 ± 2.03 mL/g·VS·d^{-1}, a 50.90 % increase over the control (27.76 ± 1.23 mL/g·VS·d^{-1}); (2) NBW shortened the microbial adaptation phase: the lag time (λ) of T3-NBW treatment was significantly reduced to 0.57 days, indicating the improved economic efficiency of the anaerobic system.

3.3. Impact of micro/nanobubbles aeration on hydrolysis/acidification

The hydrolysis process involves the conversion of complex organic compounds (such as polysaccharides, proteins, and lipids) into smaller soluble molecules, which is recognized as the rate-limiting step for methane production. As shown in Fig. 3(a), the soluble polysaccharide profile decreased until day 9 due to microbial uptake, surged from days 9–13 via enhanced hydrolysis, and finally declined as methanogenesis

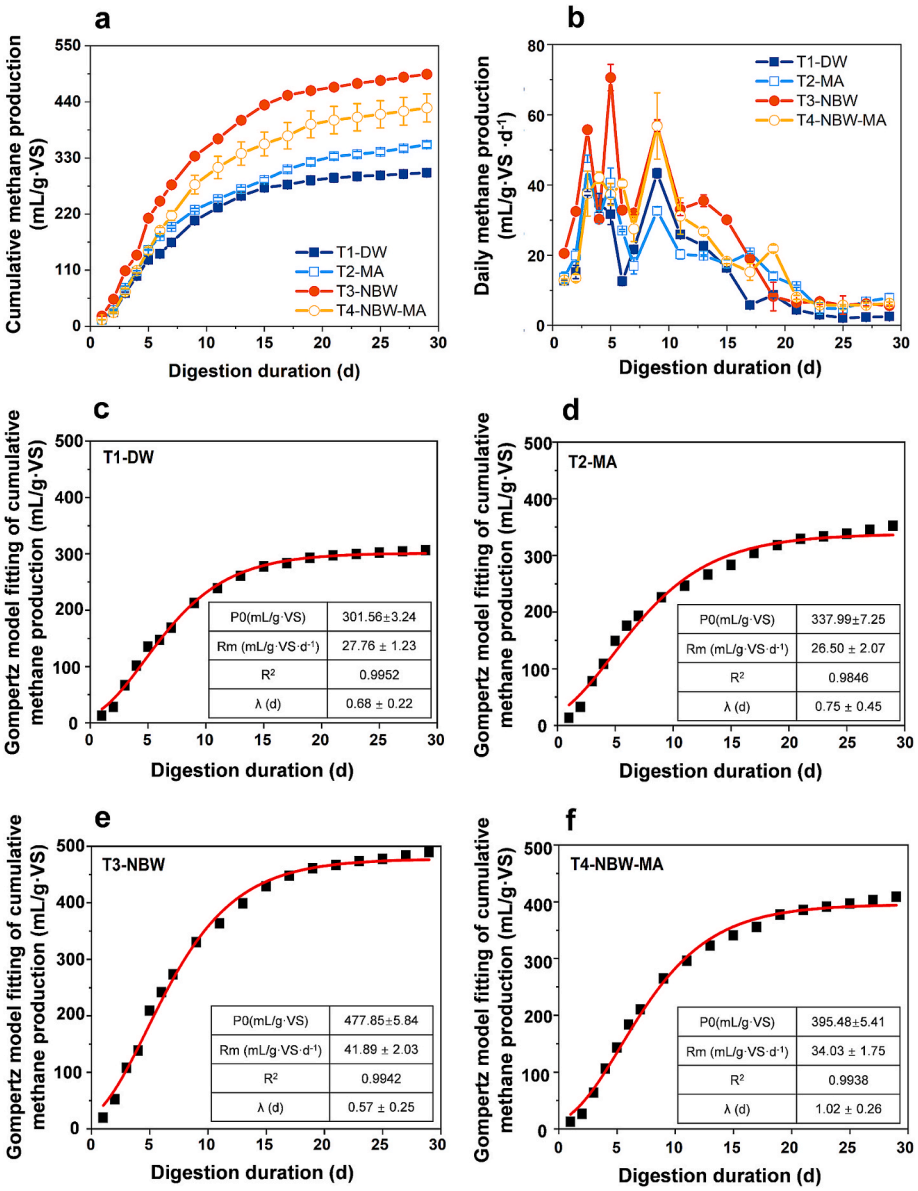


Fig. 2. Cumulative (a) and daily (b) methane production, and kinetic parameters estimated from experimental data fitted to a modified Gompertz model, T1-DW (c), d: T2-MA (d), T3-NBW (e), and T4-NBW-MA (f).

dominated. This V-shaped trajectory conformed to the typical batch operation. On day 13, the polysaccharide concentration in the T3-NBW treatment (346.68 mg/L) had increased by 13.14–24.64 % compared with the other three treatments. (278.14–305.80 mg/L). This phenomenon was also in agreement with the previous work (Zhu & Lyu, 2023a). In addition, the soluble protein concentration followed a comparable trend of first increasing and decreasing across all four treatments (Fig. 3b). The highest soluble protein concentration was reached at day 9. Compared with the control, the soluble protein concentration in the T2-MA, T3-NBW, and T4-NBW-MA treatments increased by 8.07 %, 34.12 %, and 10.92 %, respectively. Furthermore, the high specific surface area and sufficient interfacial area of NB increased opportunities for contact with microorganisms, thereby promoting rapid interaction and utilization of soluble polysaccharides and soluble proteins, and ultimately accelerating their release (Hou et al., 2024). This might explain the enhanced release of soluble polysaccharides and soluble proteins in the T3-NBW treatment. The results showed that the highest concentrations of soluble protein and polysaccharides in the T3-NBW treatment highlight its superior efficacy in hydrolyzing complex substrates. Additionally, the T3-NBW treatment exhibited the highest COD concentration, indicating the facilitated soluble organic matter released from the straw and manure. Moreover, the COD concentration in the T3-NBW treatment increased at a significantly faster rate than in other treatments from day 4 to day 17, reaching a peak on day 17 that was 21.91 % to 35.85 % higher than the others (Supplementary Material). The superior performance of the NBW treatment during the hydrolysis stage was likely attributed to the smaller size and higher stability of nanobubbles, which enabled sustained oxygen release and localized charge formation (as shown in the results of Section 3.1). Overall, these results indicated that the T3-NBW treatment significantly enhanced the solubilization of organic matter, further supporting its effectiveness in promoting substrate hydrolysis.

Macromolecular organic compounds are further decomposed into VFAs during AD, which are key intermediates for methane production. As depicted in Fig. 3(c–f), acetic and butyric acid emerged as the predominant metabolites, accompanied by relatively minor concentrations of valeric acid, propionic acid, and formic acid. Acetotrophic methanogenesis was identified as the dominant pathway. Among these, a

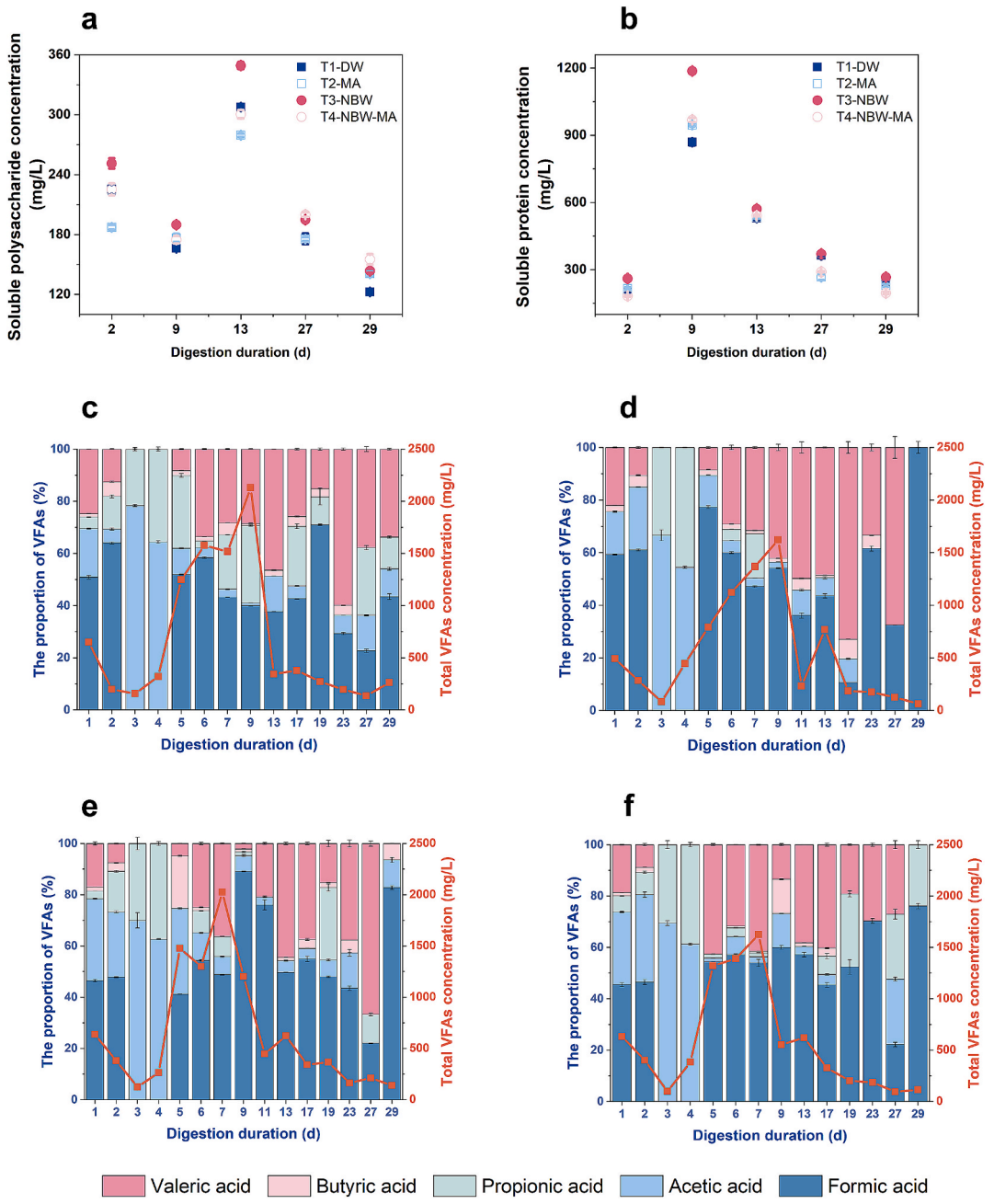


Fig. 3. Profiles of soluble protein (a) and polysaccharide (b); VFAs distribution and total VFAs in T1-DW (c), T2-MA (d), T3-NBW (e), and T4-NBW-MA treatment (f).

small amount of propionic acid existing in all treatments was conducive to the stability of the digestion system. During the first 4 days, acetic acid was the main product while butyric acid became the predominant product afterward. Butyric acid can be converted into acetic acid or other intermediate products, further promoting methane production (Wang et al., 2021). The concentrations of total VFAs (including valeric acid, butyric acid, propionic acid, acetic acid, and formic acid) remained low (<645.59 mg/L) during the early and late stages of AD in all treatments. The consistently low concentration of VFAs in the anaerobic reactor strongly indicated efficient methanogenic performance. VFAs accumulation (556.10–2136.09 mg/L) mainly occurred during 5–9 d, which was due to the hydrolysis of the complex substrate. Interestingly, it was clear that the control (T1-DW) showed relatively higher total VFAs than other treatments (T2-MA, T3-NBW, and T4-MA-NBW). This suggested that the micro-aeration strategies promoted the syntrophic relationship between the acidogens with methanogens, thereby improving the utilization of VFAs in time.

The production of butyric acid and acetic acid (BA + AA) in each treatment was present in Fig. 3(c–f). In the first 2 days, the total concentration of BA + AA in the T2-MA, T3-NBW, and T4-MA-NBW treatments ranged from 72.98 % to 85.27 %, which was slightly higher than that in the control (T1-DW). This indicated that the micro-oxygen conditions in these treatments promoted the generation of butyric and acetic acid. Notably, on day 9, the combined concentration of BA + AA in the T3-NBW treatment reached 95.13 mg/L, which was significantly higher than the 40.91 mg/L observed in the control. The elevated concentration of butyric and acetic acid implied the availability of substrates during the methanogenesis phase, which could subsequently support the growth and metabolism of methanogen. The result indicated that the addition of NBW significantly improved the degradation of organic matter and enhanced the release of soluble compounds, which subsequently enhanced methane production.

3.4. Microbial community analysis

The dynamic variation of the microbial diversity under distinct micro-aeration strategies was also investigated (Supplementary Material). Notably, T3-NBW consistently exhibited the highest Chao1 and Shannon indices, indicating enhanced species richness and diversity. For example, on day 27, the Chao1 index in the T3-NBW treatment was 823.03, which was higher than that in other treatments (659.77–745.66). This increased biodiversity contributed to greater system stability and correlated positively with methane yield (Fig. 2). It was inferred that the larger specific surface area of the NB enhanced the contact area between the bubbles and microorganisms, thereby improving the performance of AD. Prior studies also suggested that NBs facilitate microbial attachment through stronger electrostatic interactions (Hou et al., 2024). The microbial community structure of bacteria and archaea at the genus level in the four treatments was presented in Fig. 4(a–b). The dominant bacterial genera were *Psychrobacter*, *unidentified_Synergistaceae*, and *Christensenellaceae_R-7_group*. For

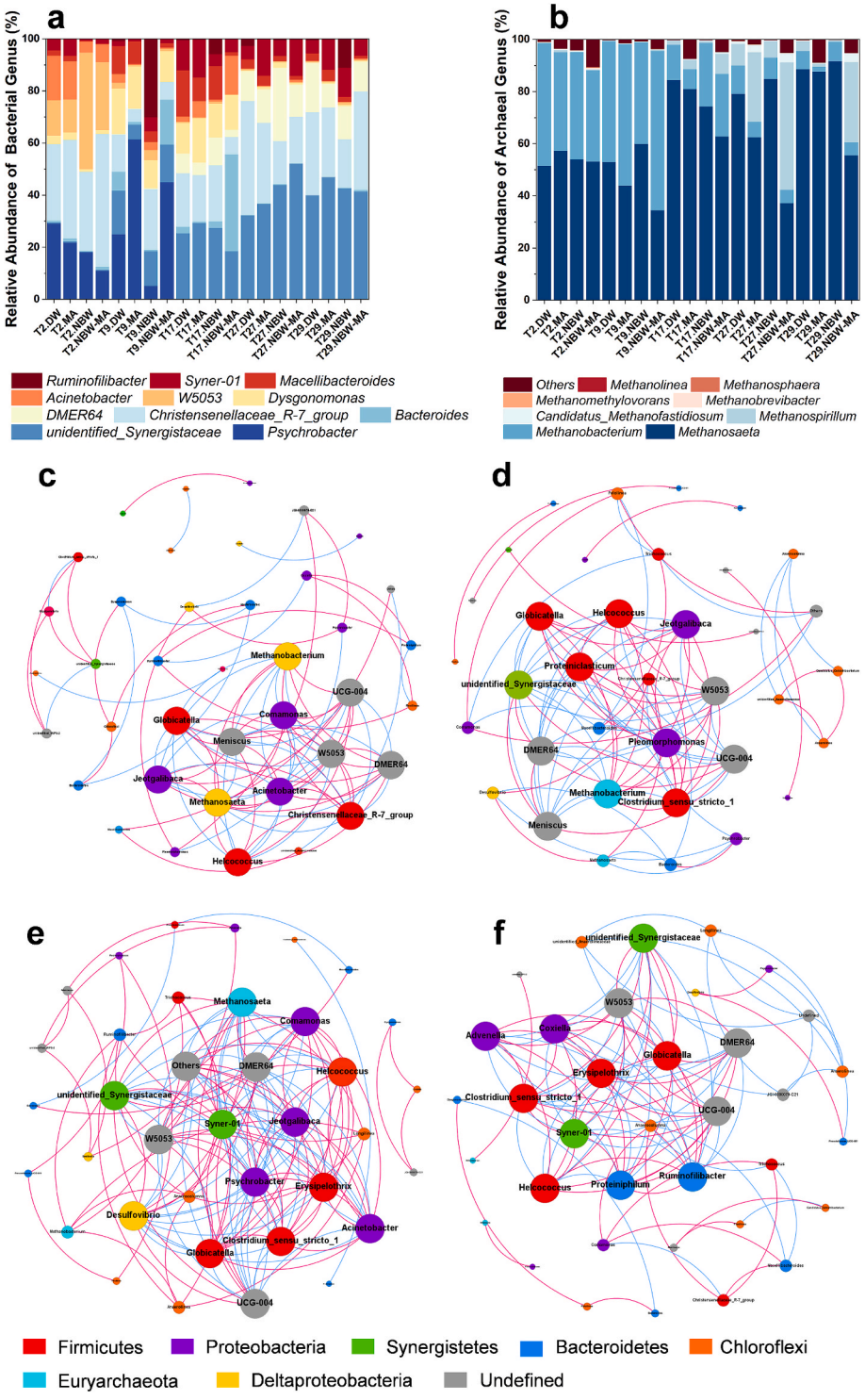


Fig. 4. The microbial community of bacterial (a) and archaeal (b); and co-occurrence network analysis of treatments T1-DW (c), T2-MA (d), T3-NBW (e), and T4-NBW-MA (f).

archaea, *Methanosaeta* and *Methanobacterium* were the dominant genera.

The *Christensenellaceae_R-7_group* was related to acetic and butyric acid production (Morotomi et al., 2012). On day 2, T3-NBW (30.36 %), T2-MA (37.87 %), and T4-NBW-MA (50.83 %) exhibited higher relative abundances of *Christensenellaceae_R-7_group* than the control (29.29 %), which increased by 3.65 %, 29.28 %, and 73.48 %, respectively. This was consistent with the higher concentrations of acetic and butyric acids in these treatments, indicating a dose-dependent relationship between gas input and the relative abundance of *Christensenellaceae_R-7_group*, which can promote VFAs production. On day 9, *Psychrobacter*—a predominantly aerobic or facultatively anaerobic genus adapted to low temperatures (Bakermans, 2018)—transiently dominated in T2-MA (61.46 %) and T4-NBW-MA (44.99 %), likely due to air/oxygen supplementation that promoted the degradation of complex substrates (e.g., proteins, lipids) into acetate and isovalerate, thereby supporting syntrophic interactions with methanogens in the AD system. As fermentation progressed, unidentified_Synergistaceae became increasingly dominant in all treatments during the later stages (days 17, 27, and 29). Their relative abundances ranged from 19.15 % to 52.31 %. This dominance was likely due to their ability to degrade proteins and carbohydrates into acetate and isovalerate. They can also supply electrons for methane production when easily degradable substrates are limited (Tomazetto et al., 2018), thereby contributing to their ecological advantage at this stage. Additionally, in the T3-NBW treatment, the abundances of *DMER64* and *Ruminofilibacter* increased by 9.45 % and 3.81–7.97 %, respectively, compared to the control. *DMER64* may enhance the degradation of disaccharides (Zhuang et al., 2021), while *Ruminofilibacter* can promote the degradation of cellulose and hemicellulose (Tan et al., 2025). These increases in abundance suggested that NBW, with its high surface area and stability, had facilitated microbial attachment and enhanced the hydrolysis of complex carbohydrates.

Methanosaeta and *Methanobacterium* are key participants in methane production during AD. *Methanobacterium*, a methane-producing bacterium with some tolerance to oxygen (Fu et al., 2016), showed relative abundances of 34.98–61.19 % during days 2–9 across all treatments. Among these, T4-NBW-MA (61.19 %) and T2-MA (54.39 %) treatments exhibited increases of 31.89 % and 17.06 %, respectively, compared to the control. This enhancement was likely due to elevated gas levels in the T2-MA and T4-NBW-MA treatments, which facilitated *Methanobacterium* growth. *Methanosaeta* is a strict acetoclastic methanogen that produces methane from acetate and releases carbon dioxide as a byproduct, playing a key role in AD (Ma et al., 2021). In this study, *Methanosaeta* dominated in all treatments on days 17, 27, and 29, with relative abundances ranging from 37.29 % to 91.66 %. Given the limited acetate content in the VFAs, it was inferred that *Methanosaeta* utilized acetate as a substrate for acetotrophic methanogenesis. Notably, T3-NBW significantly enriched *Methanosaeta*, achieving a relative abundance of 91.66 %. This indicated that the addition of NBW promoted the growth of *Methanosaeta*, thereby enhancing the acetoclastic methanogenesis process.

Co-occurrence network analysis was employed to investigate the correlations among the top 40 dominant bacterial genera across four treatments, aiming to elucidate the potential interactions between microorganisms (Fig. 4(c–f)). The network relationships of the four treatments had 97 (T1-DW), 97 (T2-MA), 151 (T3-NBW), and 109 (T4-NBW-MA) edges, respectively. Notably, the network of the T3-NBW treatment was more complex. Additionally, the proportion of positive interactions and symbiotic relationships in T3-NBW (58.28 %) was higher than that in the other three treatments. The addition of NBW enhanced microbial interactions and stabilized the ecological network in AD by promoting more intimate cooperation among microbes.

Unidentified_Synergistaceae emerged as one of the dominant bacterial genera in the T3-NBW treatment, significantly accelerating the hydrolysis and acidification of complex organic matter. This genus exhibited positive co-occurrence with 5 microorganisms, including *Methanosaeta* and *DMER64*. *Methanosaeta* is an acetoclastic methanogen that utilizes acetate to produce methane (Hong et al., 2024). The positive correlation between *Methanosaeta* and *Unidentified_Synergistaceae* suggested that NBW had enhanced the effective utilization of acetate by improving the solubility of organic matter and optimizing mass transfer processes, thereby promoting microbial synergy. Notably, *Psychrobacter* demonstrated a positive co-existence with 9 distinct microbial species in the T3-NBW treatment. In contrast, only 1–2 similar co-existence relationships were observed in the other treatments (T1-DW, T2-MA, T4-NBW-MA). Given the high specific surface area of NBW, the rupture of NBs could enhance the diffusion of substrates and metabolic products. This process enabled *Psychrobacter* to share substrates or metabolic intermediates with a wider array of microorganisms, thereby further strengthening interactions within microbial communities. For instance, the co-occurrence between *Psychrobacter* and *Acinetobacter* (a genus known for degrading aromatic compounds) may be attributed to the promotion of hydrophobic substrate dispersion by nanobubbles. Additionally, NBW can effectively alleviate oxidative stress, promoting the coexistence and cooperative metabolism between strict anaerobes and facultative anaerobes.

3.5. Functional enzymes and the proposed metabolic pathway

3.5.1. Key functional enzymes

β-glucosidase and Coenzyme F_{420} are two key enzymes involved in methane production from AD of biomass, respectively. Among them, β-glucosidase plays a pivotal role in the hydrolysis of lignocellulosic biomass, specifically in the conversion of cellulose disaccharides into glucose (Park et al., 2015). Compared to the control treatment, the activity of β-glucosidase in three treatments (T2-MA, T3-NBW, T4-NBW-MA) was enhanced to varying degrees (Supplementary Material). Notably, the T3-NBW treatment exhibited the highest increment of β-glucosidase activity over other treatments: 44.14 %, 64.94 %, 82.72 %, and 67.02 % on days 2, 9, 17, and 27, respectively. The enhanced enzyme activity could result in an improved hydrolysis process. This result also coincided with the highest methane production in T3-NBW treatment. Moreover, protease activity under NBW treatment remained consistently higher than that of the control throughout the digestion process, with a peak of 698.24 U/mL in the mid-stage—55.37 % higher than the control (449.41 U/mL). This significant enhancement further demonstrated the positive role of NBW in facilitating the hydrolysis process. Coenzyme F_{420}, a specific enzyme that plays a crucial role in methanogenesis, can serve as an indicator of methanogen activity within AD processes (Hou et al., 2021b; Wang et al., 2019). The T3-NBW treatment significantly increased coenzyme F_{420} activity by 53.17 %–133.80 % compared to the control. On day 13, the activity of coenzyme F_{420} in the T3-NBW treatment reached 28.10 U/mL, representing a 133.77 % increase compared to the control. This was consistent with previous findings that NBW addition enhanced coenzyme F_{420} activity in straw AD (Cui et al., 2024). On one hand, NBs possess high surface charge characteristics, which can enhance the catalytic efficiency of hydrolytic enzymes through adsorption-mediated mass transfer effects. On the other hand, their large specific surface area and hydrophobicity provide additional attachment sites for microorganisms, facilitating metabolic processes, accelerating organic matter degradation, and ultimately improving methane production (Niu et al., 2014; Wang et al., 2019). Thus, the addition of air-NBW enhanced the activity of key enzymes in the hydrolysis and methanogenesis stages of straw, thereby further boosting methane production.

3.5.2. Proposed metabolic pathways

Based on the metabolic functions from the KEGG database, the metabolic pathway for methane production during AD after the addition of air-NBW was proposed (Fig. 5c). Initially, during the hydrolysis phase, complex organic molecules (e.g., cellulose and glucose) are degraded by microorganisms into simpler compounds, such as monosaccharides. In the T3-NBW treatment, the activity of endoglucanases

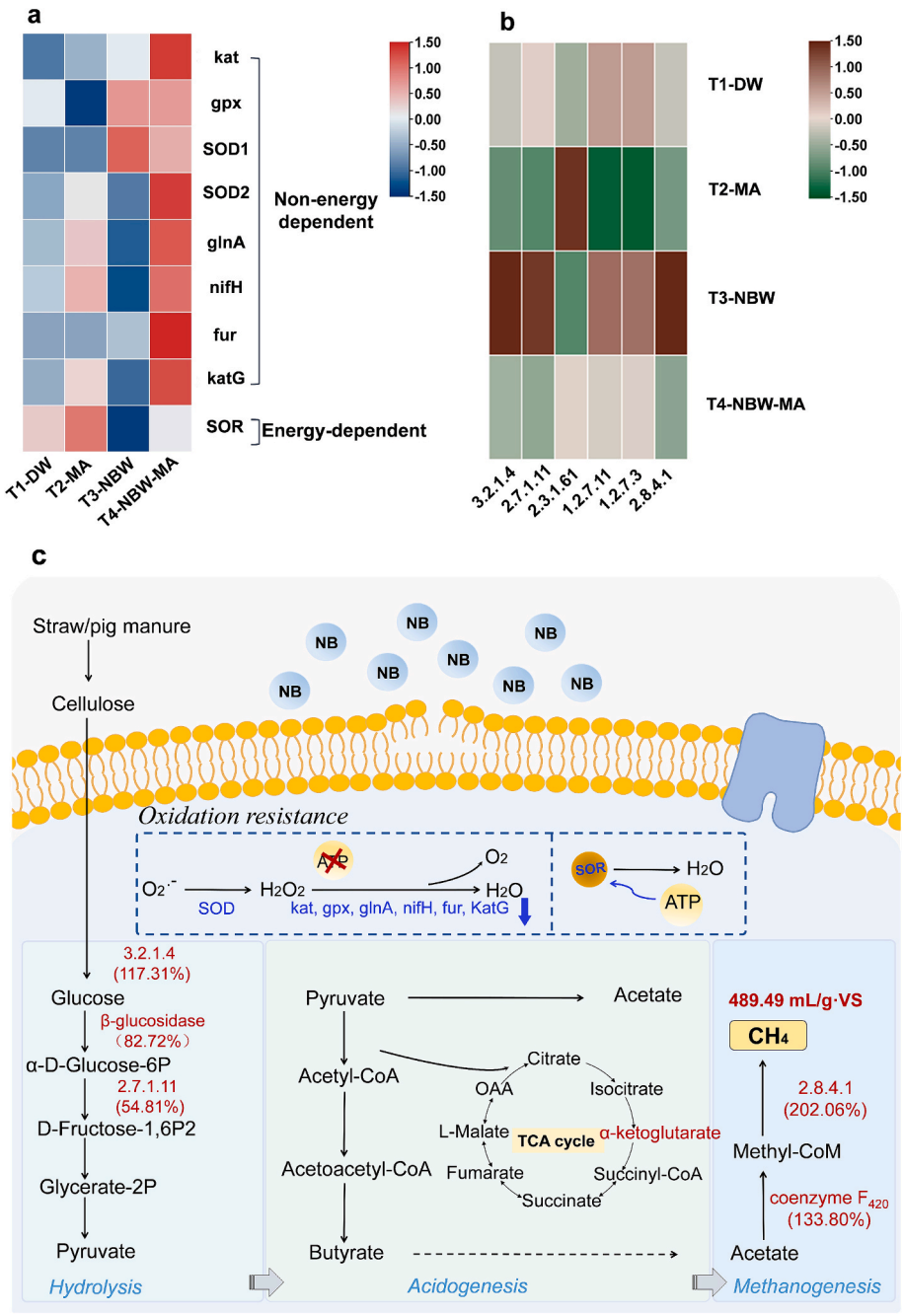


Fig. 5. Relative abundance of antioxidant enzyme-related genes (kat, gpx, SOD1, SOD2, glnA, nifH, fur, KatG, SOR) (a), and other key functional enzyme-related genes (b)during methane production, and the proposed metabolic pathways based on the KEGG pathway database (c).

(EC 3.2.1.4) was 117.31 % higher than that in the control, indicating that the introduction of NBW significantly enhanced cellulose degradation through the hydrolysis of glycosidic bonds, thereby promoting the production of glucose and facilitating subsequent acid fermentation (Fu et al., 2016; Zhu et al., 2023a). The hydrolyzed monosaccharides are converted into pyruvate via the glycolytic pathway, with 6-phosphofructokinase (EC 2.7.1.11) playing a crucial role in converting glucose to pyruvate, generating ATP and NADH in the process. Notably, the activity of 6-phosphofructokinase in the T3-NBW treatment was 54.81 % higher than in the control, enhancing sugar metabolism and energy production. Subsequently, under low oxygen conditions, pyruvate is decarboxylated to acetyl-CoA by the pyruvate dehydrogenase complex, entering the tricarboxylic acid (TCA) cycle, where it undergoes further oxidation to CO_2 and water, releasing energy and producing NADH and FADH2 (Chen et al., 2020). The activity of α-ketoglutarate dehydrogenase complex (EC 2.3.1.61, 1.2.7.11, 1.2.7.3), a key enzyme in the TCA cycle, was also elevated in the NBW treatment. The enhanced surface charge and zeta potential of NBW facilitated microbial adsorption and aggregation, which in turn promoted the accumulation of intermediate products during both acidogenesis and methanogenesis stages. In the methanogenic phase, Methyl-coenzyme M (EC 2.8.4.1) reductase activity in the T3-NBW treatment was 202.06 % higher than in the control, contributing to the increased methane yield by catalyzing the final step of methane production from methyl-CoM. These results indicated that the addition of NBW enhanced methane production by upregulating the metabolic functional genes involved in the hydrolysis, acidogenesis, and methanogenesis stages of AD.

The cavitation effect of MBs/NBs induces ROS generation (Hu et al.,

2023), activating the antioxidant defense system (such as superoxide dismutase/SOD) to maintain redox balance (Wei et al., 2020). This research investigated the expression of antioxidant enzymes under different treatments based on the KEGG database (Fig. 5a). The results showed that the expression of energy-independent antioxidant enzyme-related genes (kat, gpx, SOD1, SOD2, ginA, nifH, KatG) in the T3-NBW treatment was significantly lower than in the other three treatments. This indicated that NBs could effectively maintain appropriate ROS levels and ensure system stability, but these enzymes typically generated oxygen as a byproduct. In contrast, the T4-NBW-MA treatment exhibited the highest expression of antioxidant enzymes, reflecting that the combination of NBW and micro-aeration intensified ROS generation, prompting microorganisms to upregulate detoxification mechanisms (Liu et al., 2016a). Notably, energy-dependent enzymes such as SOR in methanogens can also eliminate ROS without producing oxygen (Wu et al., 2021), and coenzyme F_{420} can act as an electron donor to drive ROS scavenging reactions (converting ROS into harmless molecules, such as water instead of oxygen) (Martins et al., 2019). The T3-NBW treatment not only showed lower SOR abundance but also exhibited the highest coenzyme F_{420} enzyme activity. It is estimated that coenzyme F_{420} in T3-NBW treatment may participate in ROS elimination by energy-dependent enzymes like SOR. In conclusion, the T3-NBW treatment exhibited the lowest antioxidant enzyme expression, conserving energy for microbial growth and further metabolic activity. Moreover, future studies should incorporate advanced in situ detection techniques such as Electron Spin Resonance (ESR) spectroscopy to elucidate the dynamic generation and transformation mechanisms of reactive oxygen species.

4. Conclusions

This study demonstrated that T3-NBW treatment outperforms other aeration methods in enhancing methane yield during the AcoD of rice straw and pig manure. The results showed that the addition of NBW achieved the highest methane yield (489.49 mL/g·VS), representing a 59.80 % increase compared to the control. In addition, NBW significantly enhanced the activities of key enzymes (hydrolytic and methanogenic enzymes) and promoted a metabolic pathway dominated by acetotrophic methanogenesis. Further analysis revealed that NBW facilitated the development of positive microbial interactions and symbiotic relationships. Moreover, the presence of NBW reduced the energy consumption required by energy-dependent antioxidant enzymes to cope with ROS stress, thereby contributing to improved methane production. As a green and environmentally friendly additive, NBW offered advantages such as simple operation and low cost. It effectively alleviated the hydrolysis limitations in the AcoD system of rice straw and pig manure, significantly improving organic matter conversion efficiency and energy output.

CRediT authorship contribution statement

Yanning Hou: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Chao Liu:** Writing – review & editing, Methodology. **Jianhong Wei:** Writing – review & editing, Supervision. **Wenyan Zhao:** Writing – review & editing, Visualization, Supervision, Funding acquisition, Conceptualization. **Binghua Yan:** Writing – review & editing, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2025.132955.

Data availability

Data will be made available on request.

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